species to another) could avert a significant number of deaths in the future—100 times fewer deaths over a 10-year period.

"We shouldn't think that the risk of a pandemic is over," says Dr Priya Abraham, virologist and former director, National Institute of Virology (NIV), Pune. "The risk factors for a fresh outbreak are all there and increasing—global warming, international travel, changing migration patterns of animals, human encroachment in forested areas. The more we come into contact with animals, the more likely a novel virus can make the jump from an animal to a human host."

WHAT FORM WILL DISEASE X TAKE?

While the WHO has yet to specify which virus Disease X might be, experts say it is highly likely to be a respiratory virus because survival rates for these are the highest. "All humans breathe, and our respiratory tract is the fastest route for any pathogen to infect us," says Dr Aditya S. Chowti, internal medicine specialist at the Fortis Hospital on Cunningham Road in Bengaluru.

In response to the threat of Disease X, the WHO has created an R&D blueprint for a number of priority diseases. Any one of them, if they mutate to become more contagious and deadly, could become Disease X. The current list of major zoonotic viruses that threaten us include Covid-19, Crimean-Congo hemorrhagic fever, Ebola, Marburg, Lassa fever, Middle East Respiratory Syndrome (MERS) and SARS, Nipah and henipaviral diseases, Rift Valley Fever and Zika. All these have been discovered in India as well.

There are new viruses emerging every day as well, some of whom come from the SARS-CoV-2 family which produced Covid-19 and the closely-related SARS virus. In the last week of October, Chinese scientists discovered eight never-before-seen viruses on Hainan, an island off the southern coast of China. All eight were found in rodents and had a 'high probability' of infecting humans in case of a spillover event. The findings were published in the journal Virologica Sinica, and were based on 700 samples from rodents on the island. One of the viruses found belongs to the same family as the one that gave the world Covid-19.

Citing Chinese zoonotic surveillance, Mishra says India should also be making note of even the smallest disease outbreaks and zoonotic pathogens. "Not all viruses will be deadly, but we need to map out their behaviour, mutations, and if there are signs of something worrisome, we should be ahead of the curve and be able to predict and prepare beforehand itself," he says. For example, 35 people infected with an animal virus called Langya in China were identified and reported in The New England Journal of Medicine. Even though it is not a fatal or highly contagious virus, its presence is being monitored. Langya is closely related to two other henipaviruses known to infect people—Hendra and Nipah (see box above). It can cause respiratory symptoms such as fever, cough and pneumonia. Hendra and Nipah also cause respiratory infections and can be fatal (the case fatality rate is between 40-75 per cent for Nipah).

Doctors are not worried just about the number of deaths from a virus but also the long-term impact on human health. "We predict mortality and measure success if we have saved lives," says Dr Arjun Khanna, head, Department of Pulmonary Medicine at Amrita Hospital, Faridabad. "But we aren't looking at how these disease outbreaks affect quality of life." He still has patients suffering from Long Covid, as well as those who have developed asthma, COPD (Chronic Obstructive Pulmonary Disease)



"THE RISK FACTORS FOR A FRESH OUTBREAK ARE ALL THERE.... THE MORE WE COME INTO CONTACT WITH ANIMALS, THE MORE LIKELY A NOVEL VIRUS MAKES THE JUMP FROM AN ANIMAL TO A HUMAN HOST"

DR PRIYA ABRAHAM
Virologist and former director, NIV, Pune

THE ZOONOTIC VIRUSES OMEN

Over 30 new human pathogens have been detected in the last three decades, 75 per cent of which originated in animals. According to WHO, these are the ones on the watchlist. None of these have a guaranteed treatment, and only some have vaccines for prevention:

COVID-19 The respiratory illness has claimed 770 mn lives globally so far

CRIMEAN-CONGO HEMORRHAGIC FEVER Tick-borne virus causes life-threatening hemorrhagic fever. Case fatality rate (CFR) of 10-40%

EBOLA VIRUS DISEASE AND MARBURG VIRUS DISEASE Both caused by viruses of the same family, can set off hemorrhagic fevers. CFR: 24-90%

LASSA FEVER Spread by the common African rat, it is an acute viral infection, high fever for 1-4 weeks. CFR: 17%

MIDDLE EAST RESPIRATORY SYNDROME (MERS) AND SARS Same virus family, can cause severe respiratory disease. MERS has CFR of 35-40%, for SARS it is 10-13%